

Mutual Fund Analytics Platform Capstone Project

**Developed By:
Srinivasan**

**Technologies Used:
Python | SQL | SQLite | Power BI**

Bluestock Fintech



OVERVIEW

Objective:

To analyze mutual fund industry data and build an interactive analytics dashboard.

Key Focus Areas:

- Fund Performance Analysis
- Investor Analytics
- SIP Trends
- Industry Growth Analysis
- Business Intelligence Dashboard
- Result
- Conclusion



ABSTRACT

The Mutual Fund Analytics Platform is a data analytics and business intelligence solution developed to analyze mutual fund industry trends, fund performance, investor behavior, and SIP growth patterns. The project integrates 10 financial datasets and applies ETL, data cleaning, SQL analytics, and advanced performance metrics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, and Beta.

Interactive Power BI dashboards were developed to provide insights into industry growth, fund performance, investor transactions, and market trends. The platform transforms raw financial data into actionable insights that support data-driven investment analysis and decision-making.



PROBLEM STATEMENT

Challenges:

- Large volume of mutual fund data
- Difficult to identify top-performing funds
- Limited visibility into investor behavior
- Lack of consolidated analytics platform

Solution:

Develop a centralized Mutual Fund Analytics Platform.



DATASET OVERVIEW

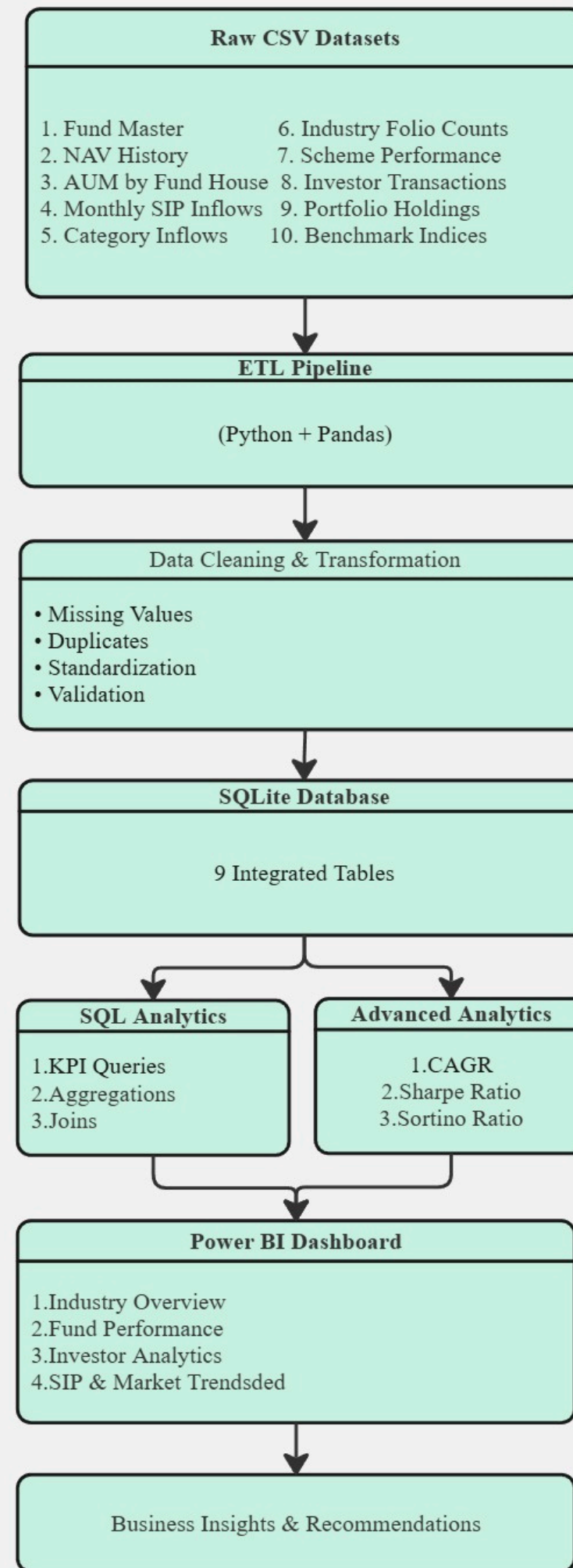
Datasets Used:

- Fund Master
- NAV History
- Performance
- Transactions
- SIP
- AUM
- Category
- Benchmark
- Folio

Total Records Processed:
87,211+



DATA ARCHITECTURE



ETL & DATA CLEANING

- Data Processing Activities:
- Missing Value Handling
- Duplicate Removal
- Standardization
- Validation Checks
- Feature Engineering
- Result:
- Clean and Analysis-Ready Dataset

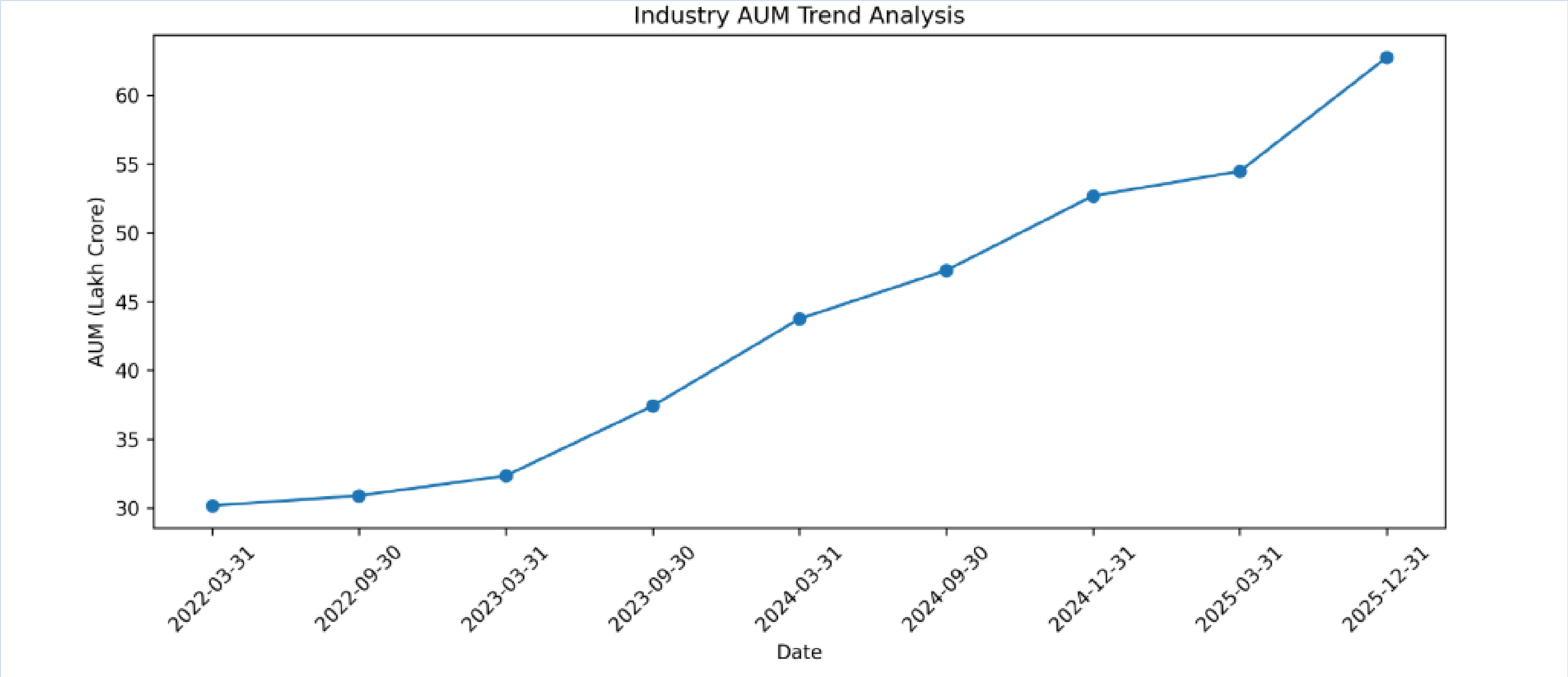


EXPLORATORY DATA ANALYSIS

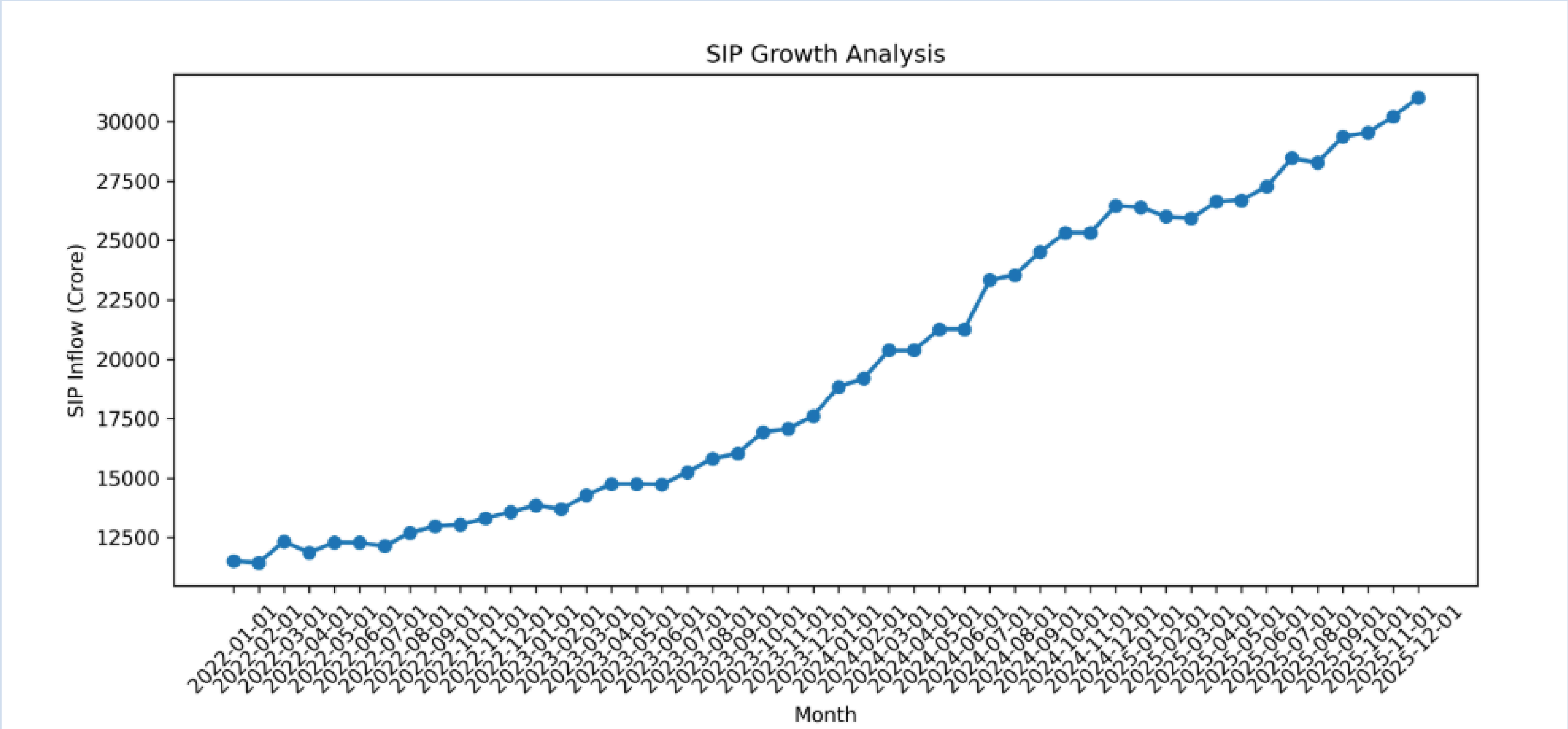
- Industry AUM increased steadily
- SIP participation showed growth
- Category inflows varied significantly
- Investor participation expanded



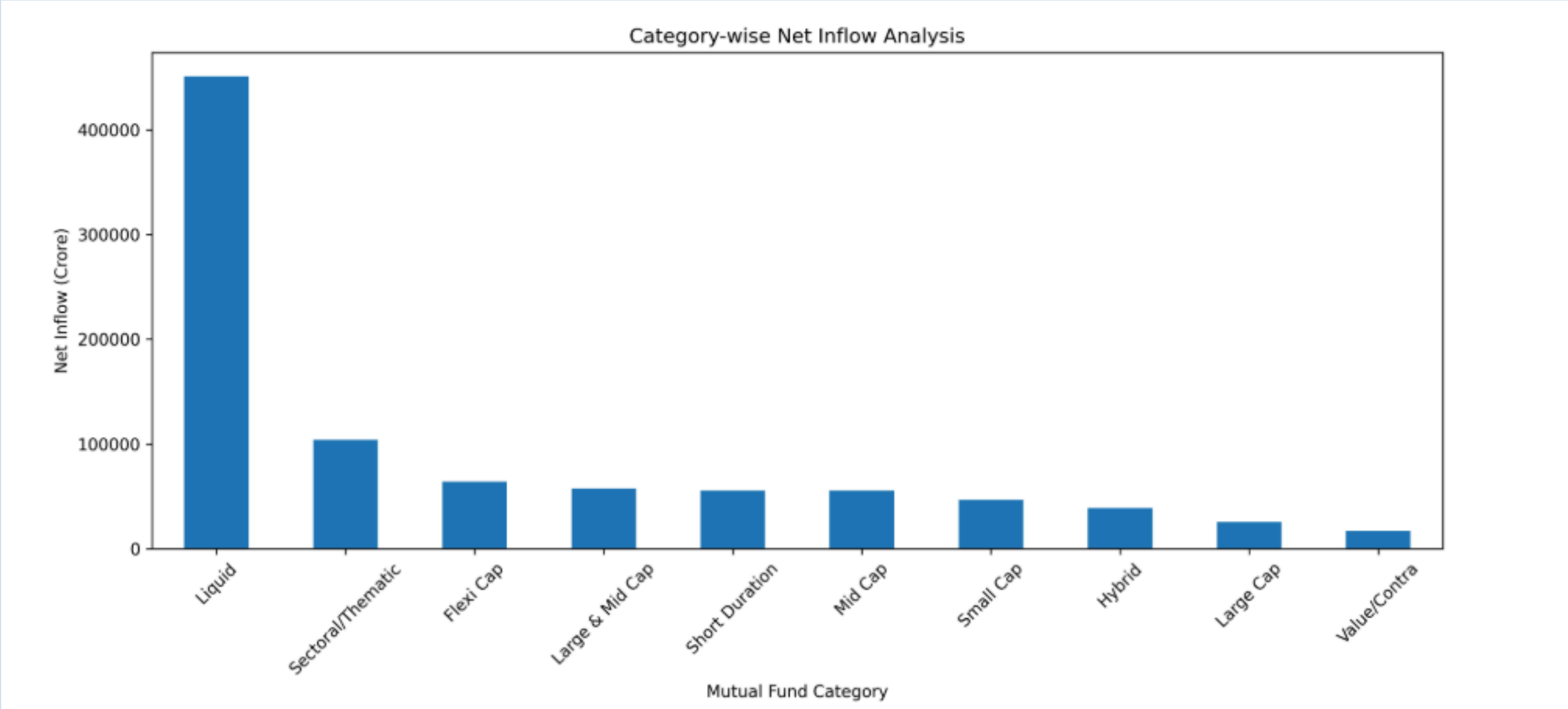
EXPLORATORY DATA ANALYSIS



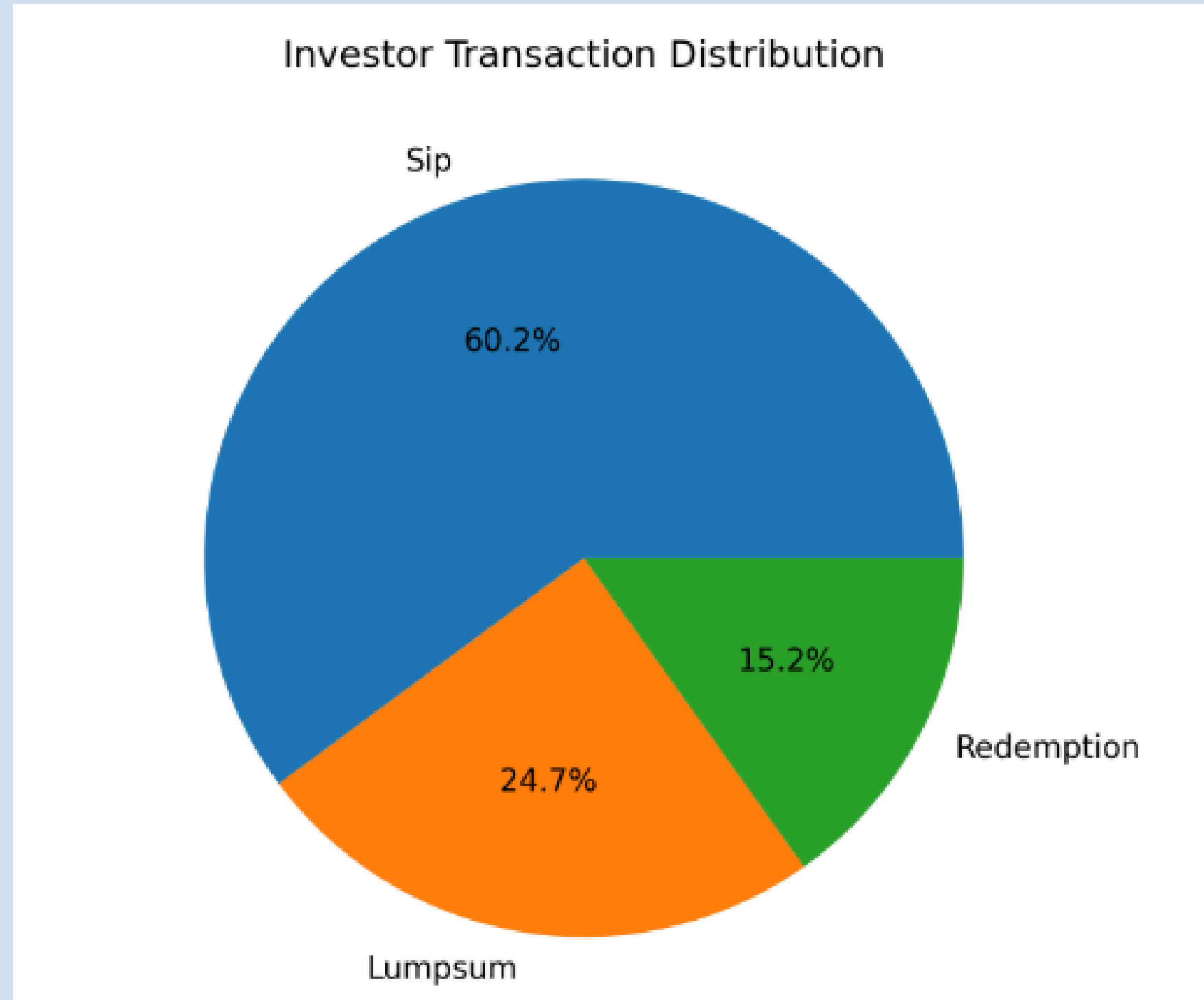
EXPLORATORY DATA ANALYSIS



EXPLORATORY DATA ANALYSIS



EXPLORATORY DATA ANALYSIS



PERFORMANCE ANALYTICS

- Industry AUM increased steadily
- SIP participation showed growth
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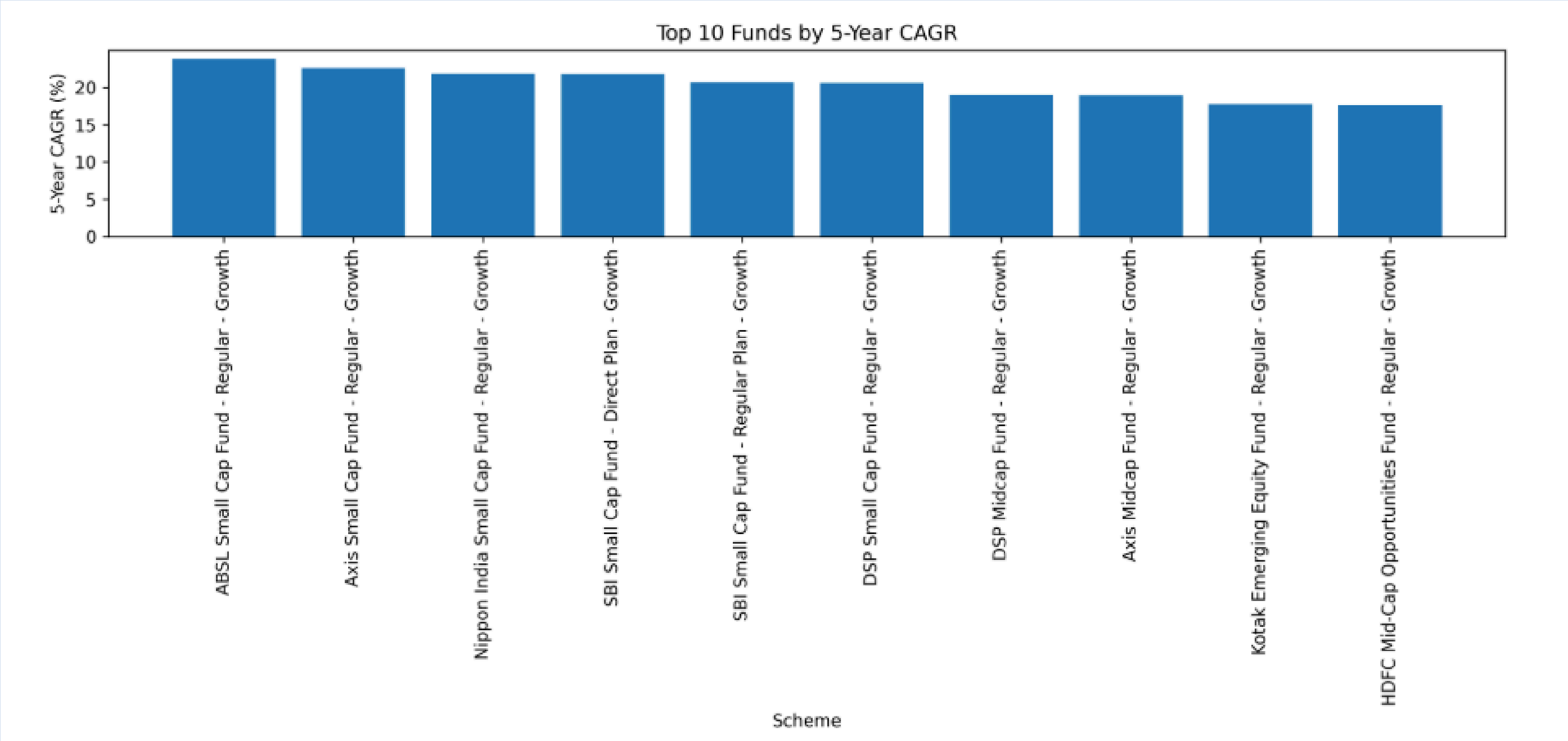
PERFORMANCE ANALYTICS

Metrics Used:

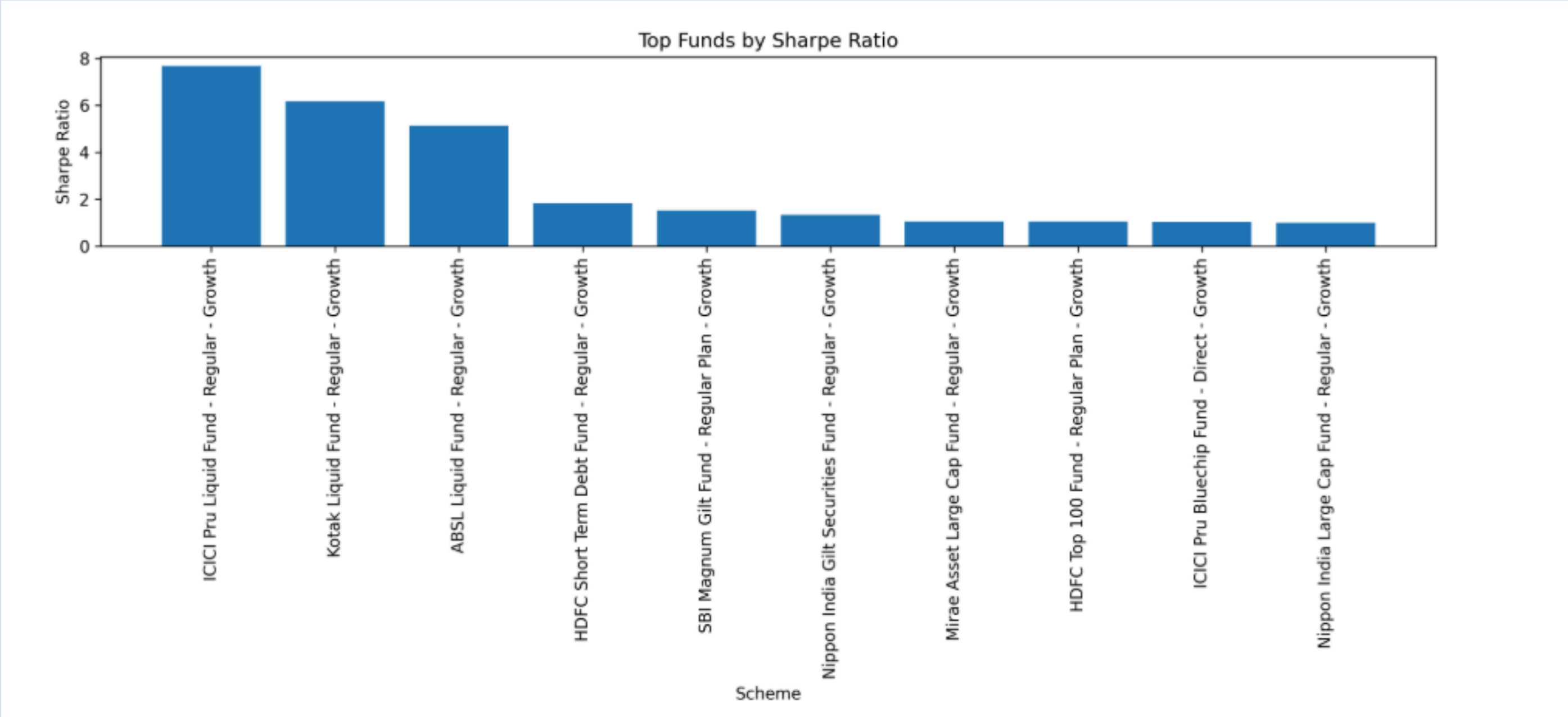
- CAGR
- Sharpe Ratio
- Sortino Ratio
- Alpha
- Beta
- Max Drawdown



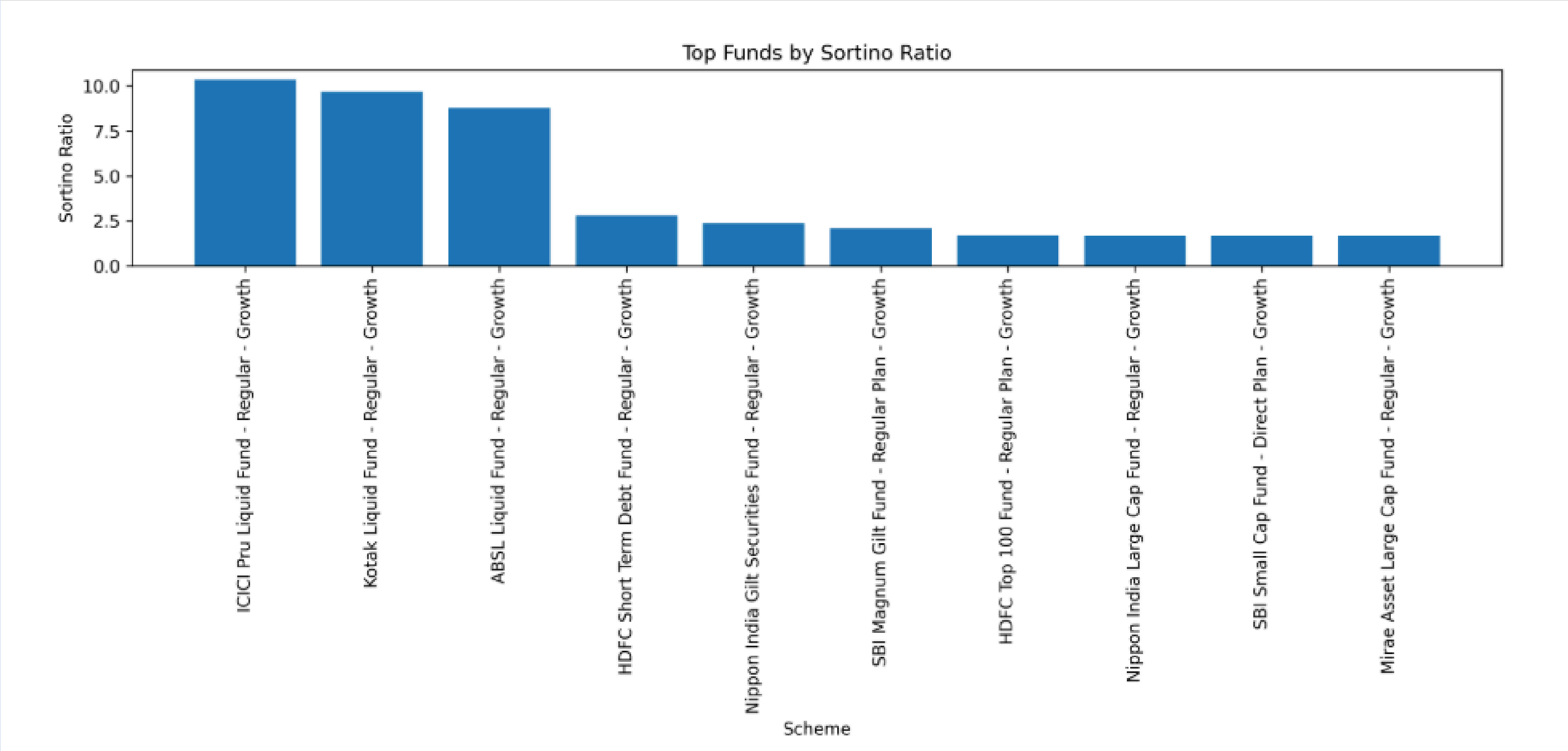
PERFORMANCE ANALYTICS



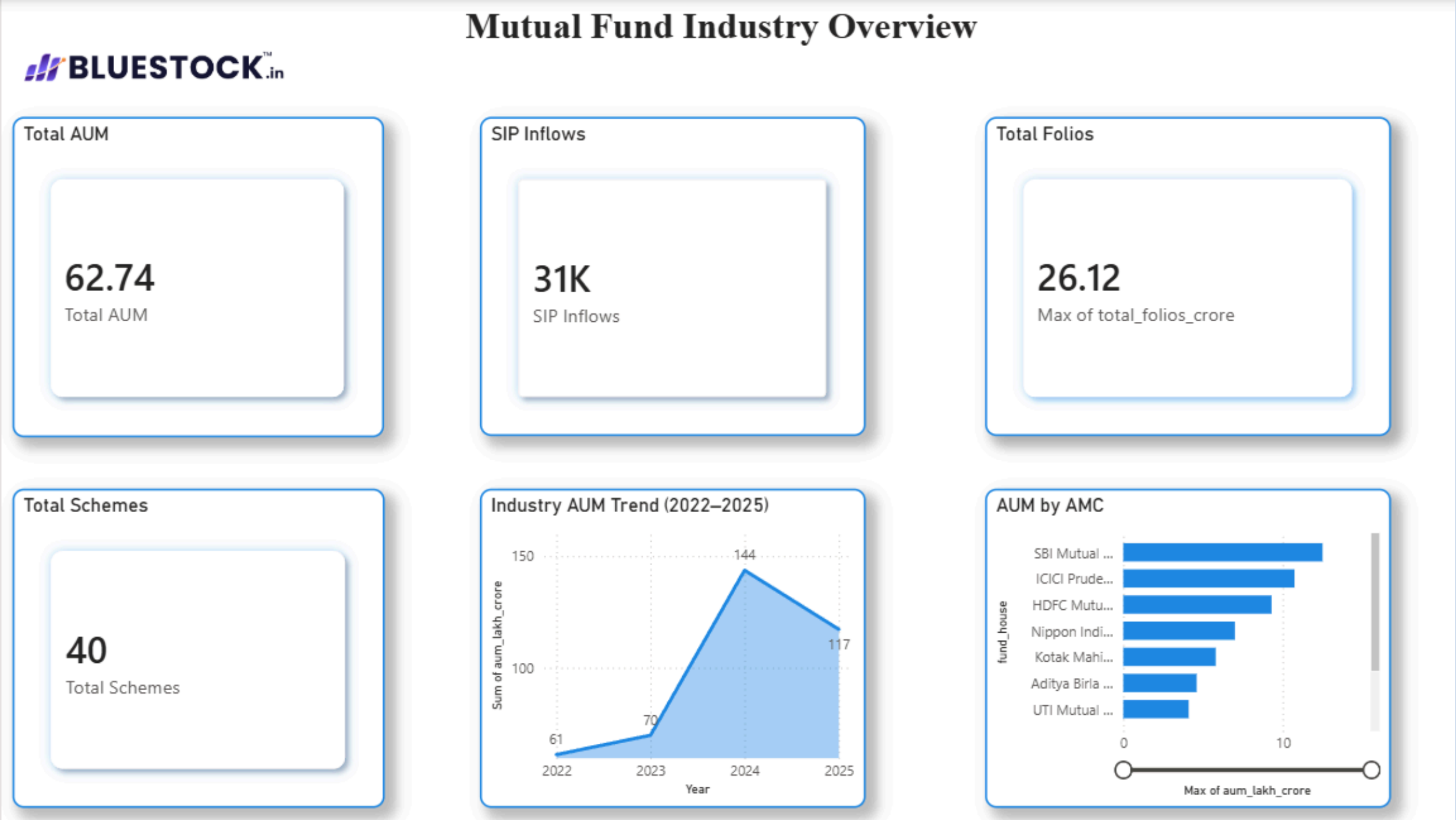
PERFORMANCE ANALYTICS



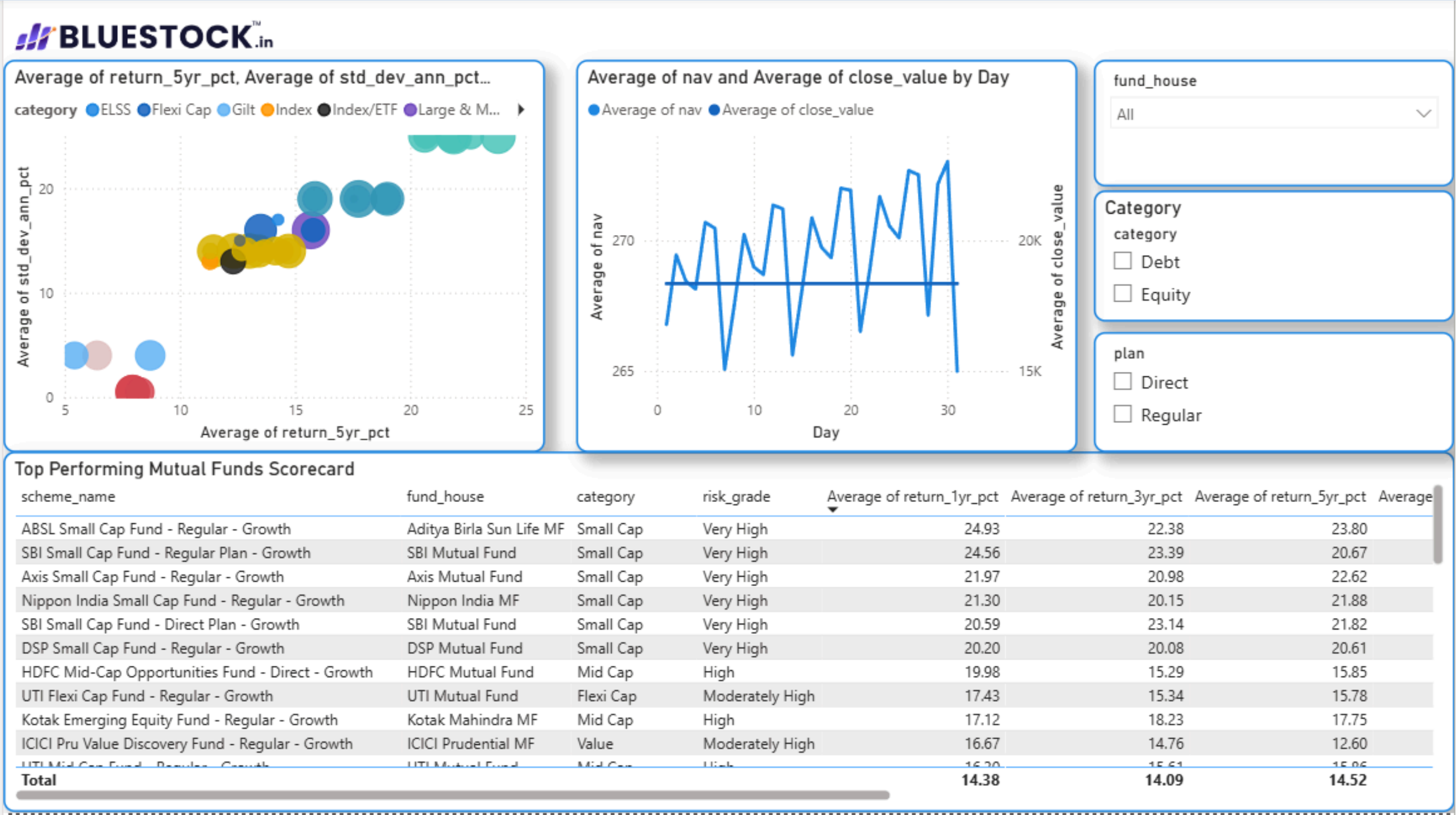
PERFORMANCE ANALYTICS



DASHBOARD INDUSTRY OVERVIEW



FUND PERFORMANCE

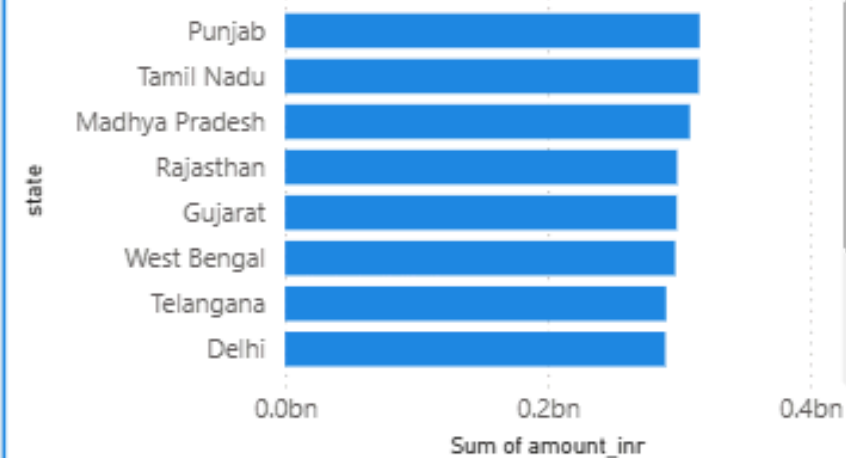


Top Performing Mutual Funds Scorecard

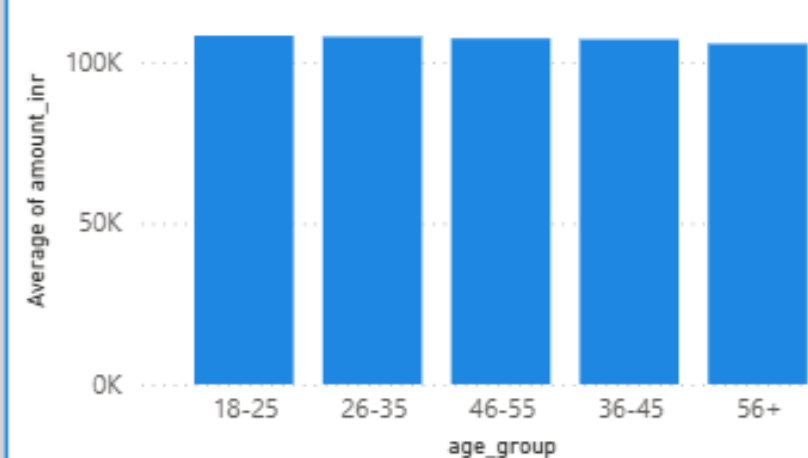
scheme_name	fund_house	category	risk_grade	Average of return_1yr_pct	Average of return_3yr_pct	Average of return_5yr_pct	Average
ABSL Small Cap Fund - Regular - Growth	Aditya Birla Sun Life MF	Small Cap	Very High	24.93	22.38	23.80	
SBI Small Cap Fund - Regular Plan - Growth	SBI Mutual Fund	Small Cap	Very High	24.56	23.39	20.67	
Axis Small Cap Fund - Regular - Growth	Axis Mutual Fund	Small Cap	Very High	21.97	20.98	22.62	
Nippon India Small Cap Fund - Regular - Growth	Nippon India MF	Small Cap	Very High	21.30	20.15	21.88	
SBI Small Cap Fund - Direct Plan - Growth	SBI Mutual Fund	Small Cap	Very High	20.59	23.14	21.82	
DSP Small Cap Fund - Regular - Growth	DSP Mutual Fund	Small Cap	Very High	20.20	20.08	20.61	
HDFC Mid-Cap Opportunities Fund - Direct - Growth	HDFC Mutual Fund	Mid Cap	High	19.98	15.29	15.85	
UTI Flexi Cap Fund - Regular - Growth	UTI Mutual Fund	Flexi Cap	Moderately High	17.43	15.34	15.78	
Kotak Emerging Equity Fund - Regular - Growth	Kotak Mahindra MF	Mid Cap	High	17.12	18.23	17.75	
ICICI Pru Value Discovery Fund - Regular - Growth	ICICI Prudential MF	Value	Moderately High	16.67	14.76	12.60	
UTI Mid Cap Fund - Regular - Growth	UTI Mutual Fund	Mid Cap	High	16.20	15.61	15.86	
Total				14.38	14.09	14.52	

INVESTOR ANALYTICS

Transaction amount by state



Average SIP Amount by Age Group



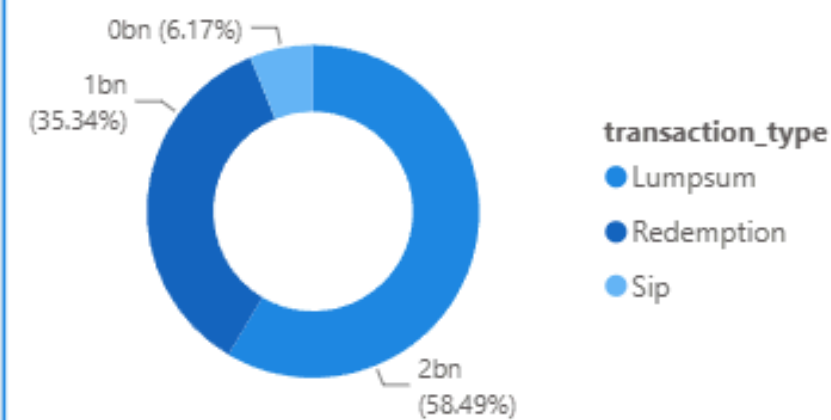
state

- ☐ Delhi
- ☐ Gujarat
- ☐ Haryana
- ☐ Karnataka
- ☐ Madhya Pradesh
- ☐ Maharashtra
- ☐ Punjab
- ☐ Rajasthan

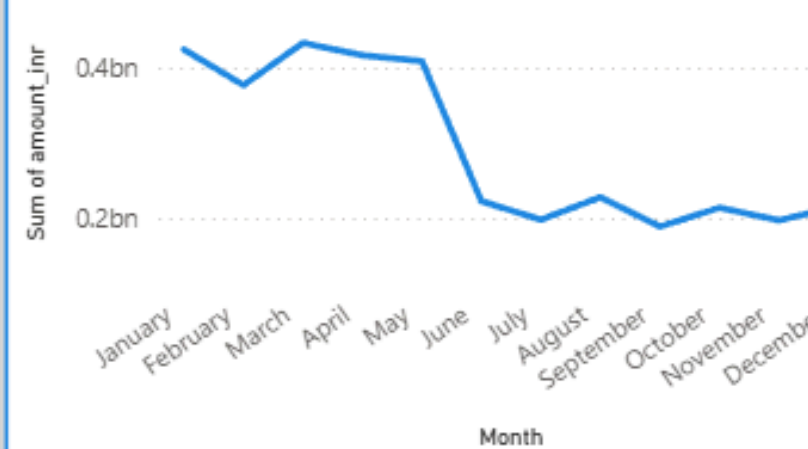
age_group

- ☐ 18-25
- ☐ 26-35
- ☐ 36-45
- ☐ 46-55
- ☐ 56+

Transaction Type Distribution



Monthly Transaction Volume Trend



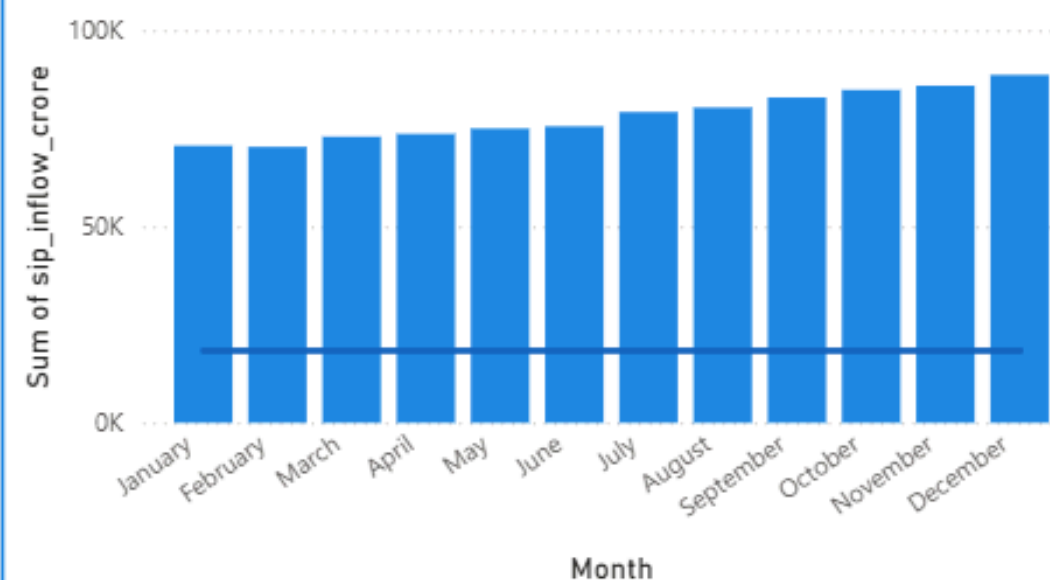
city_tier

- ☐ B30
- ☐ T30

SIP & MARKET TRENDS

SIP Inflows vs Nifty 50 (2022–2025)

● Sum of sip_inflow_crore ● Average of close_value



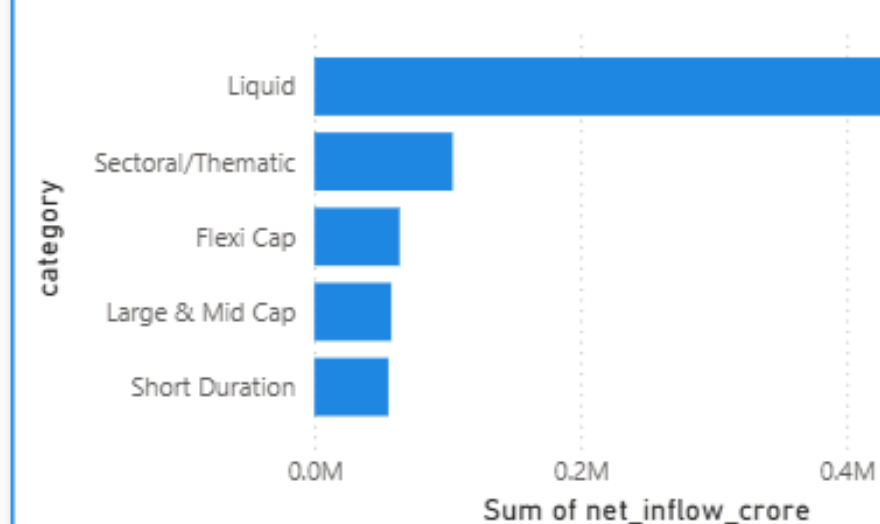
index_name

- ☐ BSE_SMALLCAP
- ☐ CRISIL_GILT
- ☐ CRISIL_LIQUID
- ☐ NIFTY_MIDCAP150
- ☐ NIFTY100
- ☐ NIFTY50
- ☐ NIFTY500

Category Inflow Heatmap.

category	2024	2025	Total
ELSS	4627	1453	6080
Flexi Cap	47551	16438	63989
Gilt	7753	2642	10395
Hybrid	29711	9157	38868
Large & Mid Cap	43169	14583	57752
Large Cap	19449	6184	25633
Liquid	346328	104947	451275
Mid Cap	41116	14196	55312
Sectoral/Thematic	78107	25722	103829
Short Duration	41859	13671	55530
Small Cap	34204	12392	46596
Value/Contra	12744	4236	16980
Total	706618	225621	932239

Top 5 Categories by Net Inflow FY25



KEY FINDINGS

- AUM growth remained strong
- SIP inflows increased consistently
- Investor participation expanded
- Top funds outperformed benchmarks
- Risk-adjusted metrics improved fund evaluation



CONCLUSION

The Mutual Fund Analytics Platform successfully transformed raw financial data into actionable business insights using Python, SQL, SQLite, and Power BI.

The solution enables efficient analysis of mutual fund performance, investor behavior, and industry trends through an interactive dashboard.



B1 – AUTOMATED ETL SCHEDULING

- Developed Python ETL pipeline to fetch NAV data from mfapi.in
- Automated data collection every weekday at 8:00 PM
- Configured using Windows Task Scheduler
- Stored latest NAV data in CSV format
- Eliminated manual data refresh process



B1 – AUTOMATED ETL SCHEDULING

1	date,nav,fetch_date
2	11-06-2026,105.57370,2026-06-12 17:35:18.584285
3	10-06-2026,105.65690,2026-06-12 17:35:18.584285
4	09-06-2026,105.57930,2026-06-12 17:35:18.584285
5	08-06-2026,105.30840,2026-06-12 17:35:18.584285
6	05-06-2026,105.10700,2026-06-12 17:35:18.584285
7	04-06-2026,104.80950,2026-06-12 17:35:18.584285
8	03-06-2026,104.74960,2026-06-12 17:35:18.584285
9	02-06-2026,104.76630,2026-06-12 17:35:18.584285
10	01-06-2026,104.70250,2026-06-12 17:35:18.584285
11	29-05-2026,104.65700,2026-06-12 17:35:18.584285
12	27-05-2026,104.58330,2026-06-12 17:35:18.584285
13	26-05-2026,104.52690,2026-06-12 17:35:18.584285

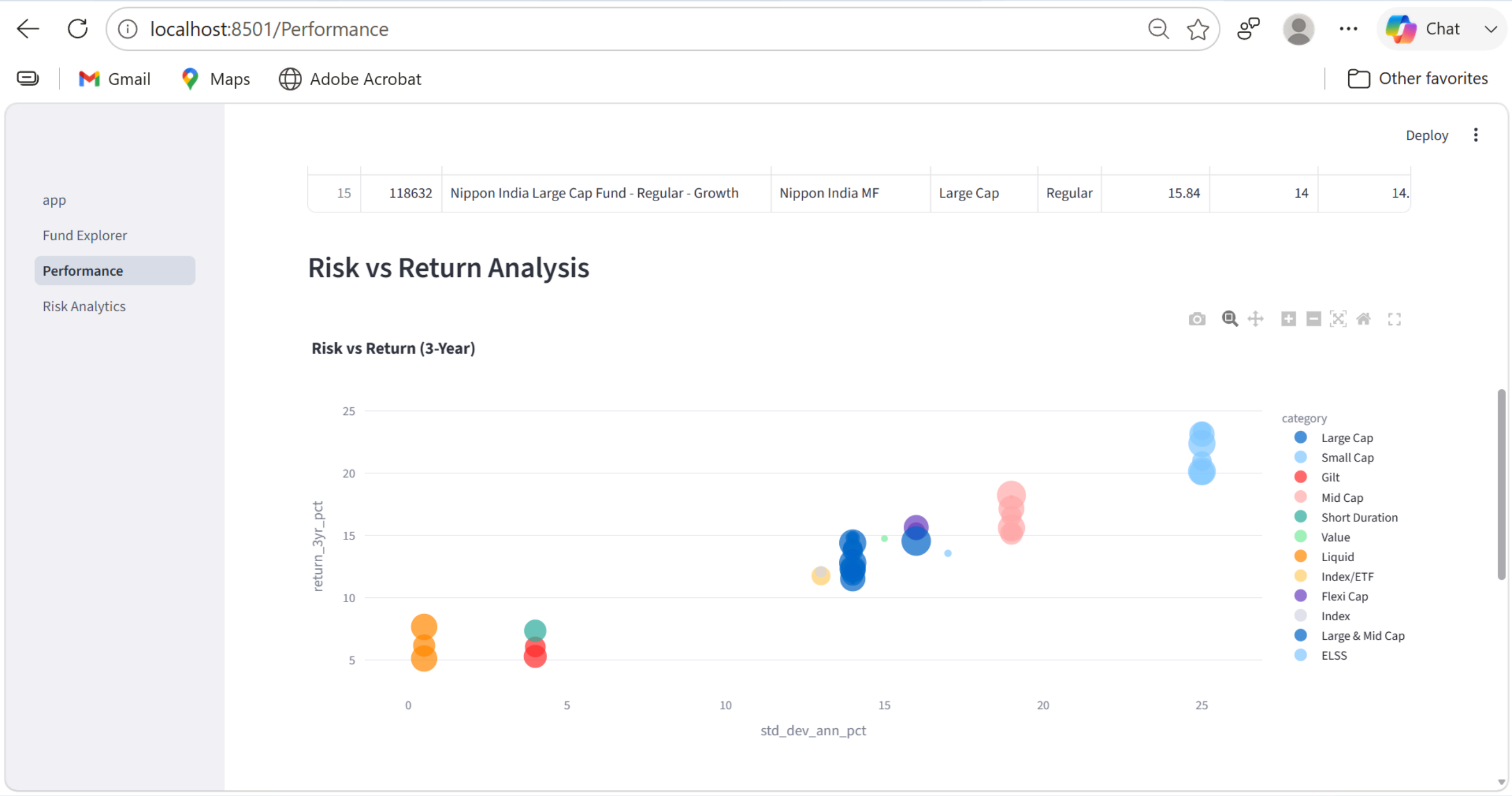


B2 – STREAMLIT ANALYTICS DASHBOARD

- Built an interactive web application using Streamlit
- Developed Industry Overview dashboard
- Added Fund Explorer and Search functionality
- Implemented Performance Analytics visualizations
- Created Risk Analytics and Recommendation Engine



B2 – STREAMLIT ANALYTICS DASHBOARD

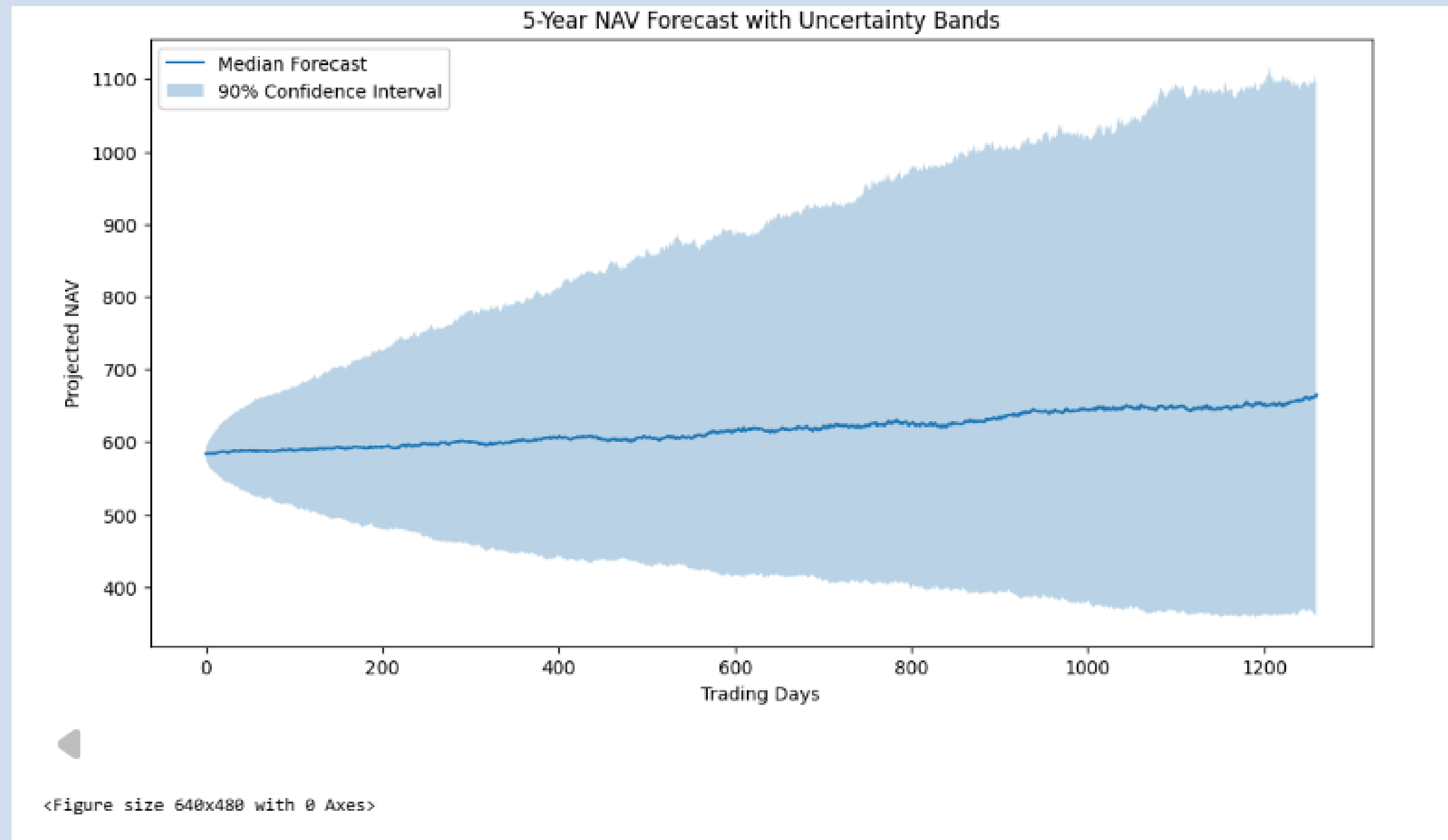


B3 – MONTE CARLO NAV FORECASTING

- Simulated future NAV growth for 5 years
- Used historical daily return distribution
- Generated 1,000 simulation scenarios
- Calculated uncertainty and confidence bands
- Visualized possible future NAV outcomes



B3 – MONTE CARLO NAV FORECASTING

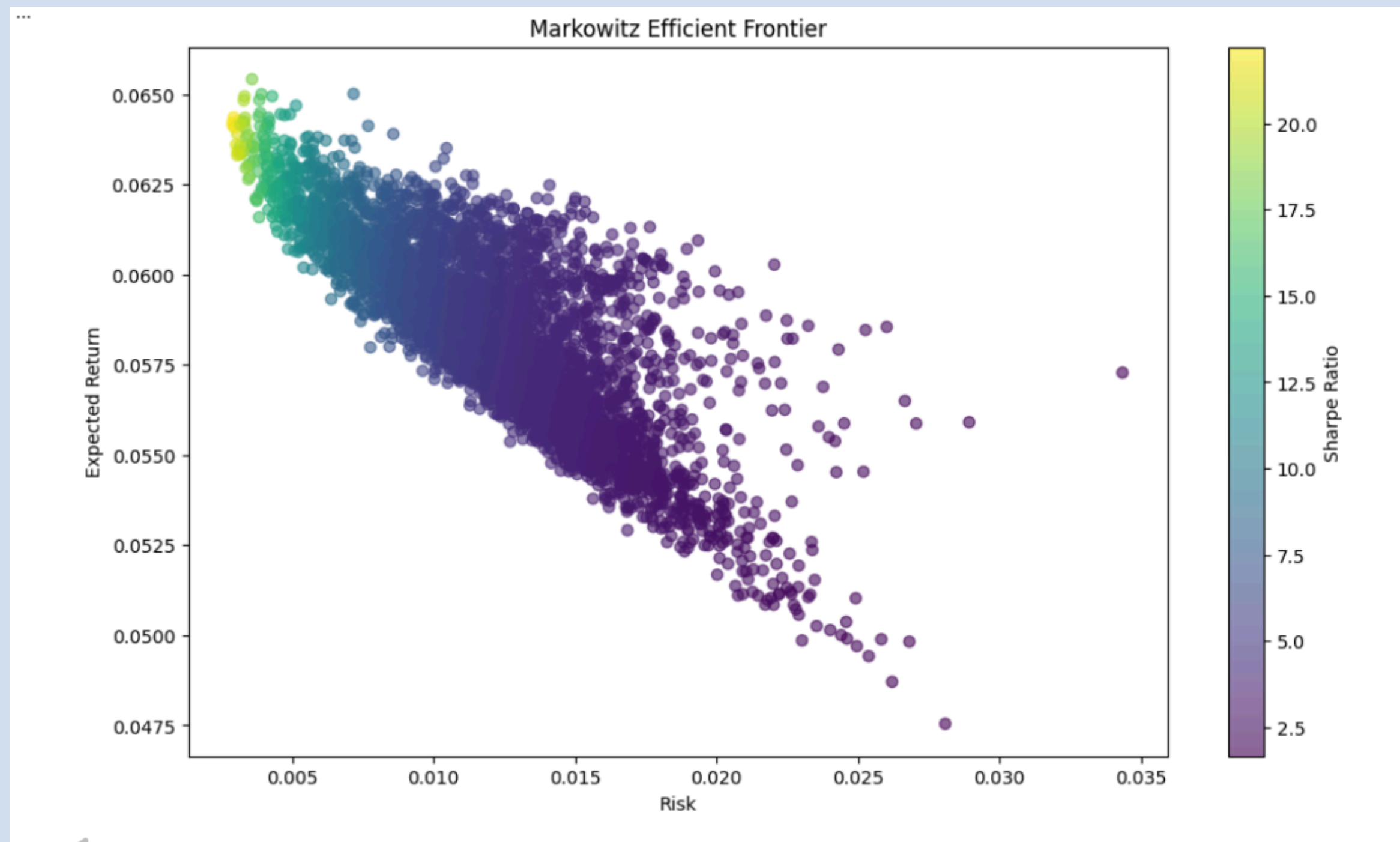


B4 – EFFICIENT FRONTIER PORTFOLIO OPTIMIZATION

- Selected 5 mutual funds for optimization
- Generated 5,000 random portfolios
- Calculated expected return and portfolio risk
- Computed Sharpe Ratio for each portfolio
- Identified optimal portfolio allocation



B4 – EFFICIENT FRONTIER PORTFOLIO OPTIMIZATION



B5 – AUTOMATED HTML EMAIL REPORTING

- Generated HTML performance reports automatically
- Retrieved data from SQLite database
- Delivered reports through Gmail SMTP
- Enabled weekly report distribution
- Reduced manual reporting effort



B5 – AUTOMATED HTML EMAIL REPORTING

Weekly Mutual Fund Performance Report

S

srinivasan2004ss@gmail.com

To: You

← ↶ ↷ ...

Fri 2026-06-12 17:44

Weekly Mutual Fund Performance Report

Top 10 Funds Based on 3-Year Returns

	scheme_name	return_3yr_pct	sharpe_ratio	aum_crore
	SBI Small Cap Fund - Regular Plan - Growth	23.39	0.94	19259
	SBI Small Cap Fund - Direct Plan - Growth	23.14	0.93	36061
	ABSL Small Cap Fund - Regular - Growth	22.38	0.90	41613
	Axis Small Cap Fund - Regular - Growth	20.98	0.84	21545
	Nippon India Small Cap Fund - Regular - Growth	20.15	0.81	43630
	DSP Small Cap Fund - Regular - Growth	20.08	0.80	35124
	Kotak Emerging Equity Fund - Regular - Growth	18.23	0.96	47469
	ICICI Pru Midcap Fund - Regular - Growth	18.08	0.95	979
	DSP Midcap Fund - Regular - Growth	17.16	0.90	37835
	HDFC Mid-Cap Opportunities Fund - Regular - Growth	16.58	0.87	23185



Thank You

